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Kristýna Kramer (Zemková)

Curriculum Vitae

Personal

E-mail	kristyna.kramer@gmail.com
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Nationality	Czech Republic
Address	Via Armando Falconi 8, 20128 Milan, Italy

Employment in Academia

06/2025 – 12/2025	**Charles University**, Czechia Postdoc
01/2024 – 08/2024	**University of Victoria**, Canada PIMS Postdoctoral Fellow Mentor: Stephen Scully
01/2023 – 12/2023	**University of Alberta**, Canada PIMS Postdoctoral Fellow Mentor: Stefan Gille
10/2018 – 12/2022	**TU Dortmund University**, Germany Scientific staff (Wissenschaftliche Mitarbeiter)

Education

2018 – 2022	**TU Dortmund University, Faculty of Mathematics**, Germany PhD Advisor: Detlev Hoffmann Thesis: *Equivalence relations of quadratic forms in characteristic 2 and quasilinear p-forms*
2015 – 2018	**Charles University, Faculty of Mathematics and Physics**, Czechia Mgr. (Master's degree), Mathematical Structures Advisor: Vítězslav Kala Thesis: *Composition of quadratic forms over number fields* (Dean's prize for the best thesis; 3rd place in SVOČ research competition) Graduated with distinction
2012 – 2015	**Charles University, Faculty of Mathematics and Physics**, Czechia Bc. (Bachelor's degree), General Mathematics Advisor: Jan Šaroch Thesis: *Norm-euclidean quadratic extensions of the field of rational numbers*

Other Employment

01/2013 – 08/2018	Cooperation with the publishing house **Didaktis** Co-writing textbooks on mathematics for high schools
03/2016 – 12/2017	Cooperation on the project **Techambition** Creating online lectures from mathematics for high school students

Study Experience

09/2024 – 03/2025	Language school **ELLCI** in Milan, Italy Italian course
09/2018	**Max-Planck Institute for Mathematics** in Bonn, Germany Internship (with Pieter Moree)
09/2016 – 02/2017	**Technical University of Darmstadt**, Germany Exchange scholarship (Erasmus+)
02/2016 – 06/2016	**Polytechnic University of Catalonia**, Spain Exchange scholarship (Erasmus+)

Publications

[12] *Non-universality of ternary quadratic forms over fields containing* $\sqrt{2}$,
(with J. Krásenský), in preparation

[11] *Kitaoka's conjecture and sums of squares*,
(with V. Kala and J. Krásenský), submitted

[10] *Vishik equivalence and similarity of quadratic forms over fields of characteristic 2,*
(with D. Hoffmann), J. Algebra, 2025

[9] *Isotropy indices of Pfister multiples in characteristic two,*
(with N. Lorenz), J. Pure Appl. Algebra, 2024

[8] *Vishik equivalence and similarity of p-forms and totally singular quadratic forms*,
Pacific J. Math., 2024

[7] *Isotropy and the full splitting pattern of quasilinear p-forms*,
J. Algebra, 2024

[6] *Isotropy of Quadratic Forms over Function Fields in Characteristic 2*
Comm. Algebra, 2024

[5] *Composition of Bhargava's cubes over number fields*,
Expo. Math., 2023

[4] *Composition of Binary Quadratic Forms over Number Fields*,
Math. Slovaca, 2021

[3] *Integers represented by ternary quadratic forms*
(with B. Faye, L. Matthiesen, D. Schindler and M. Tinková),
Women in Numbers Europe III: Research Directions in Number Theory, 2020

[2] *There are no universal ternary quadratic forms over biquadratic fields*
(with J. Krásenský and M. Tinková), Proc. Edinburgh Math. Soc., 2020

[1] *Universal Quadratic Forms and Indecomposables over Biquadratic Fields*
(with M. Čech, D. Lachman, J. Svoboda, M. Tinková), Math. Nachr., 2018

Grants

01/2023 – 08/2024	PIMS Postdoctoral Fellowship Two-year postdoctoral funding awarded by Pacific Institute for the Mathematical Sciences
01/2018 – 06/2018	GAUK (Grant Agency for Charles University) Research grant, co-investigator

	Indecomposable elements and universal quadratic forms
11/2017 – 05/2018	Charles University, Faculty of Mathematics and Physics Faculty grant for students, principal investigator *Indecomposable elements in biquadratic number fields*
05/2017 – 11/2017	Charles University, Faculty of Mathematics and Physics Faculty grant for students, principal investigator *Universal quadratic forms over number fields*

Conferences and Talks

25 – 29/08/2025	ALGAR 2025: Algebras with involution and linear algebraic groups University of Antwerp, Belgium
24/10/2024	Al@Bicocca Seminar Università degli Studi di Milano-Bicocca, Milan, Italy Presentation: *Equivalence relations of quadratic forms in characteristic two*
31/05 – 03/06/2024	Canadian Mathematical Society Summer Meeting University of Saskatchewan, Saskatoon, Canada
08 – 13/10/2023	Motives and Invariants: Theory and Applications to Algebraic Groups and their Torsors BIRS, Banff, Canada
11 – 15/09/2023	Young Number Theorists in Bonn Max Planck Institute for Mathematics, Bonn, Germany
21 – 25/08/2023	ALGAR 2023: Local-global principles for quadratic forms University of Antwerp, Belgium
03/05/2023	The PIMS Postdoctoral Fellow Seminar Presentation: *Isotropy of quadratic forms in characteristic 2*
23/11/2022	Number Theory Lunch Seminar Max Planck Institute for Mathematics, Bonn, Germany Presentation: *Equivalence relations of quadratic forms, mainly in characteristic two*
12 – 16/09/2022	Motives in Ratisbona University of Regensburg, Germany
05 – 09/09/2022	ALGAR 2022: Constructive methods for quadratic form theory University of Antwerp, Belgium Presentation: *Quadratic forms in characteristic two*
05/07/2022	University of Antwerp, Belgium Presentation: *Isotropy of Quadratic Forms over Function Fields*
25/11/2021	Oberseminar Algebra und Geometrie TU Dortmund, Germany (online) Presentation: *Equivalence relations of quadratic forms in characteristic two*
04 – 07/11/2021	Autumn School of Algebra (organized by the Department of Algebra at the Charles University) Presentation: *Isotropy of quadratic forms in characteristic two*
16 – 20/08/2021	ALGAR 2021: Systems of quadratic forms over arithmetic fields University of Antwerp, Belgium

29/03/2021	Guest lecture on quadratic forms (online) Charles University Topic: *Quadratic forms in the parallel universe where 1+1=0*
06 – 10/07/2020	ALGAR 2020: Valuations, quadratic forms and definability University of Antwerp, Belgium (online)
26 – 30/08/2019	Women in Numbers Europe 3, Rennes, France Project: *Integers represented by ternary quadratic forms*
01-05/07/2019	ALGAR 2019: Algebras with involution University of Antwerp, Belgium
22/11/2018	Oberseminar Algebra und Geometrie TU Dortmund, Germany Presentation: *Composition of quadratic forms over number fields*
03 – 07/09/2018	ELAZ (Elementare und Analytische Zahlentheorie) Max Planck Institute for Mathematics, Bonn, Germany
02 – 06/07/2018	ALGAR 2018: Sums of squares in fields University of Antwerp, Belgium
09 – 12/11/2017	Autumn School of Algebra (organized by the Department of Algebra at the Charles University) Presentation: *Universal quadratic forms over number fields*
17/10/2017	Number Theory Seminar Charles University, Czech Republic Presentation: *Composition of quadratic forms over number fields*
17 – 23/09/2017	20th International Workshop for Young Mathematicians "Number Theory" Kraków, Poland Presentation: *Composition of binary quadratic forms*
04 – 07/09/2017	International Conference on Class Groups of Number Fields & Related Topics Allahabad, India
03 – 07/07/2017	XXXth Journées Arithmétiques Caen, France
19 – 22/11/2015	Autumn School of Algebra (organized by the Department of Algebra at the Charles University) Presentation: *Integer partitions and the use of modular forms*
03/11/2015	Number Theory Seminar Charles University, Czech Republic Presentation: *The existence of the Euclidean norm in quadratic number fields*

Teaching

University of Victoria, Canada

- MATH 101 – Calculus II (lecturer), Spring 2024

University of Alberta, Canada

- MATH 125 – Linear Algebra (lecturer), Spring 2023

Technical University Dortmund, Germany

- Linear Algebra 1 (recitation), Winter 2022/23
- Proseminar to Linear Algebra (assistant/supervisor), Winter 2020/21 and 2021/22
- Seminar to Algebra and Number Theory (assistant/supervisor), Summer 2021 and 2022
- Quadratic Forms in Characteristic 2 – part 2 (recitation), Summer 2020
- Quadratic Forms in Characteristic 2 – part 1 (recitation), Winter 2019/20

Charles University, Czech Republic

- Linear Algebra 2 (recitation), Summer 2018

Languages

- **Czech** (Native)
- **English** (Advanced)
- **German** (Advanced)
- **Italian** (Intermediate/B1)
- **Dutch** (Elementary)

Interests

- **Cooking and baking:** I eat exclusively vegan to reduce my impact on the environment. I love to improvise and only rarely use recipes.
- **Hiking and sleeping under the sky:** Nothing gives me a bigger sense of freedom than walking through a forest alone with my backpack.
- **Riding a kickbike:** Neither running nor biking but a lot of squats. It is kind of a meditative experience, as I always count to four to change the standing leg regularly.